Mahbouba

Empowering Tunisian Women Farmers through AI, IoT, and MLOps

AGRI-HOPE Technical Challenge

Team Members

Ahmed Bensalah	Engineering Student
Emna Mnejja	Engineering Student
Nour Elhouda Benslimene	Engineering Student

IEEE WIEACT Tunisia Section

October 4, 2025

Contents

1 Introduction **2**

2 System Overview and Tools **2**
2.1 Technology Stack and Roles 2

3 Architecture **3**

4 AI Microservices and Screenshots **3**
4.1 Speech-to-Text (STT – Whisper.cpp) 3
4.2 Text-to-Speech (TTS – Coqui TTS) 4
4.3 Language Model (LLM – Labess-7B Chat) 4
4.4 NGINX API Gateway 5

5 IoT and GPT-OSS:120B Integration **5**

6 Class Diagram **6**

7 MLOps Lifecycle with GitHub Actions **6**

8 Achievements **7**

9 Conclusion **7**

Abstract

Mahbouba is a smart agricultural platform empowering Tunisian women farmers through AI, IoT, and voice-based learning. Built on Azure and Firebase, it integrates microservices for speech recognition, synthesis, and natural dialogue—connected to an IoT weather network powered by GPT-OSS:120B for predictive risk alerts. The system achieves continuous delivery and model reproducibility via GitHub Actions and DVC pipelines. This report details the full architecture, toolchain, and operational flow supporting this inclusive agritech innovation.

1 Introduction

Mahbouba merges speech AI, IoT meteorology, and MLOps automation to help women farmers anticipate risks, learn entrepreneurial skills, and connect with local markets. Its core modules enable:

- Voice-based learning in Tunisian Arabic.
- Real-time weather and soil monitoring.
- Predictive alerts for crop protection.
- Marketplace integration with buyers and suppliers.

2 System Overview and Tools

2.1 Technology Stack and Roles

- **Flutter:** Frontend framework for a cross-platform app with audio recording and playback support.
- **Firebase:** Web backend for buyers/suppliers with authentication and live Firestore database.
- **Azure Cloud:** Hosts containerized AI microservices and IoT data ingestion.
- **Redis:** In-memory cache for TTS/LLM responses.
- **DVC (Data Version Control):** Ensures traceable model evolution and dataset integrity.
- **NGINX Gateway:** API router balancing traffic among services securely.
- **GitHub Actions:** Implements CI/CD workflows—automatically builds, tests, and deploys containers from Git commits.
- **GPT-OSS:120B:** Large open model used for environmental forecasting and risk classification.
- **IoT Meteo Station:** Hardware device collecting humidity, temperature, soil, and rainfall data in real time.

Project Repository: All source code, configuration files, and Docker definitions are publicly available on GitHub: `https://github.com/EmnaMnejja6/mahbouba.git`

3 Architecture

Mahbouba follows a modular cloud architecture with three AI microservices (STT, TTS, LLM) deployed as containers. IoT data streams feed predictive models, and all interactions flow through an NGINX gateway.

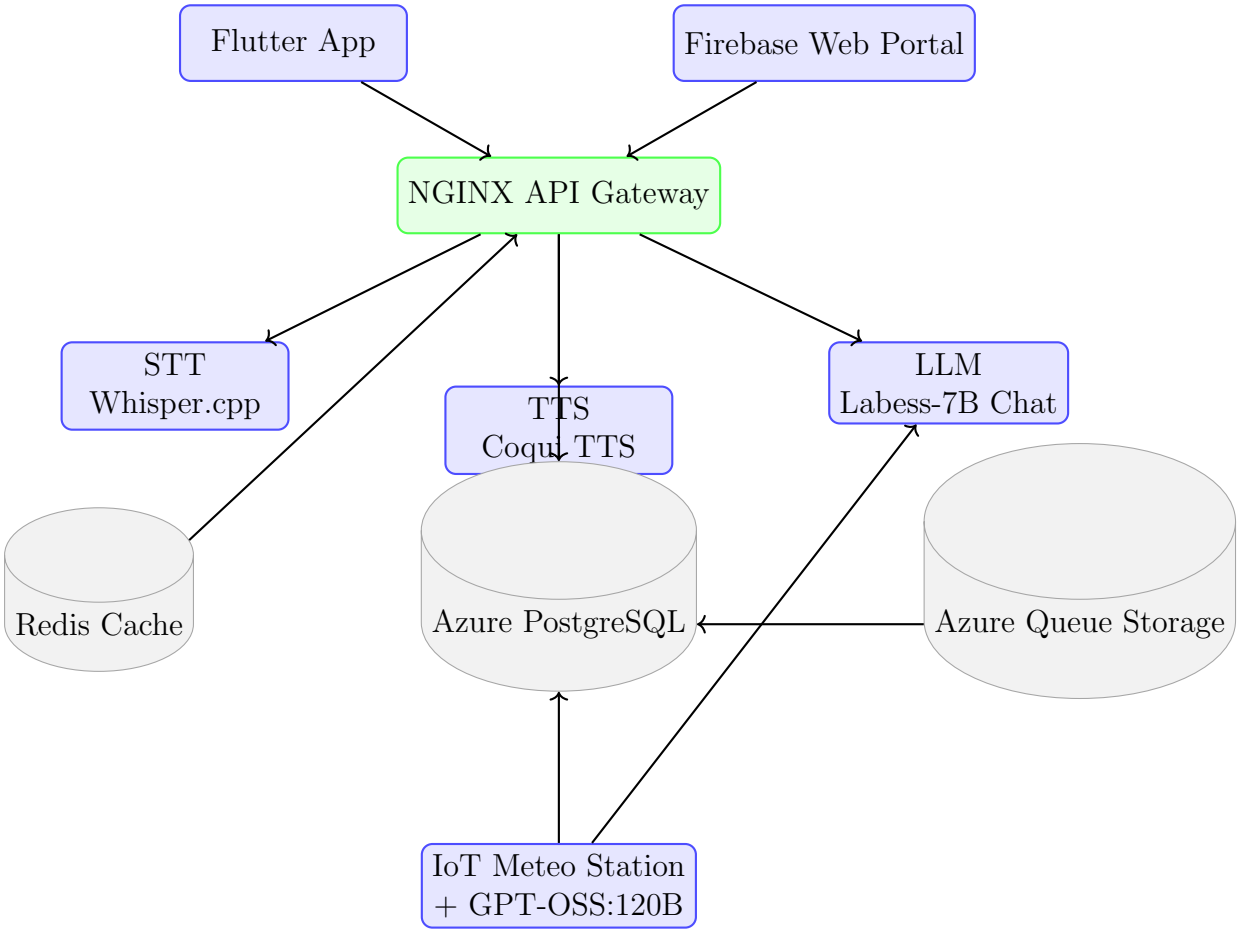


Figure 1: Mahbouba integrated architecture (AI, IoT, and MLOps).

4 AI Microservices and Screenshots

4.1 Speech-to-Text (STT – Whisper.cpp)

Fine-tuned on Tunisian Arabic voice data, Whisper.cpp converts farmers' speech to text on CPU-optimized containers. **Steps:** record → send audio → transcribe → cache result.

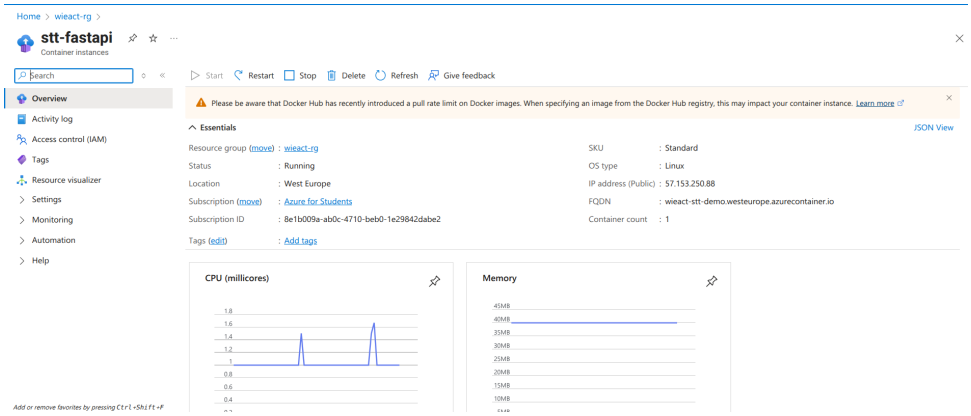


Figure 2: STT microservice console showing transcription latency and logs.

4.2 Text-to-Speech (TTS – Coqui TTS)

Coqui TTS generates natural Tunisian voices. It serves both educational content and weather alerts via cached audio streams.

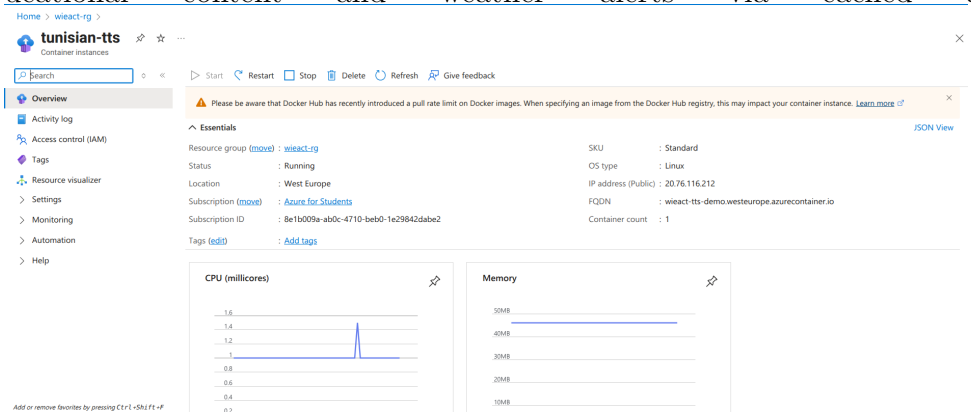


Figure 3: Coqui TTS dashboard during Arabic model inference.

4.3 Language Model (LLM – Labess-7B Chat)

The Labess-7B-Chat model interprets text, answers questions, and creates responses localized for Tunisian entrepreneurship.

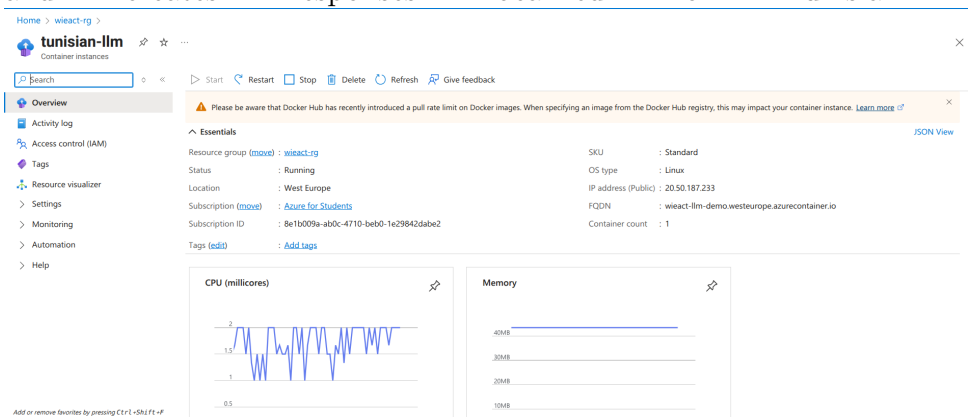


Figure 4: LLM interaction log visualized on Azure Container Monitor.

4.4 NGINX API Gateway

Manages unified access to all backend services, enforces HTTPS, and routes REST/JSON requests.

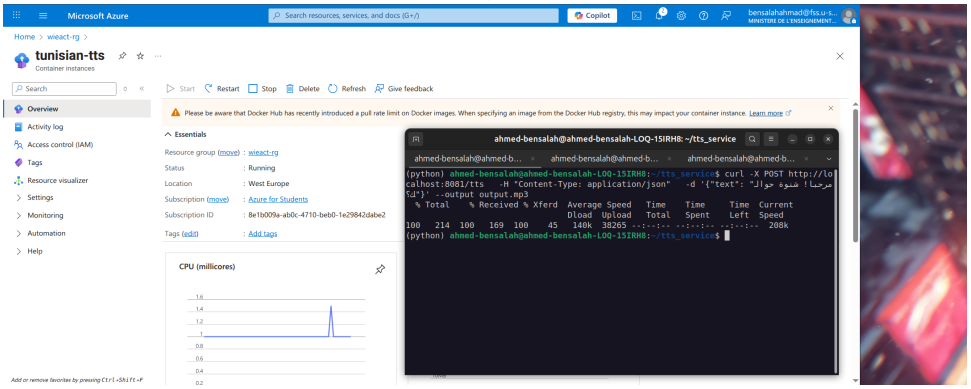


Figure 5: NGINX Gateway routing logs with service health checks.

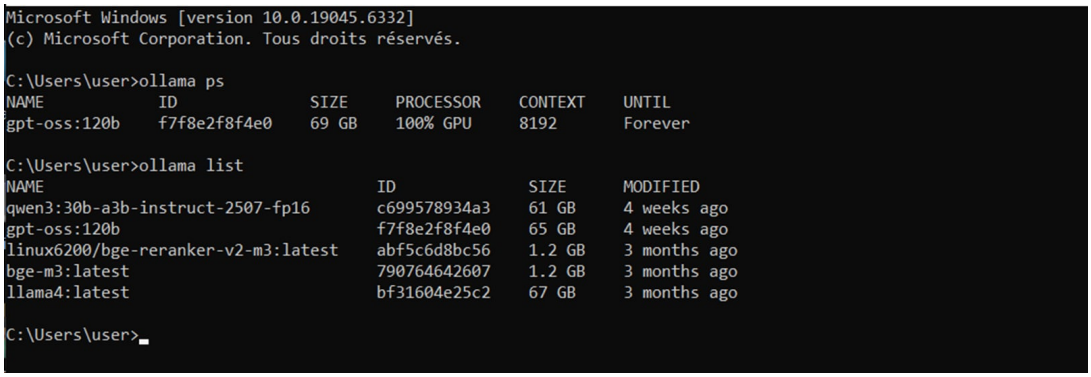
5 IoT and GPT-OSS:120B Integration

The IoT station collects real-time metrics:

- Temperature, humidity, rainfall, soil moisture, and UV radiation.
- Data transmitted via LoRaWAN to Azure IoT Hub.

GPT-OSS:120B analyzes these signals to:

- Predict environmental stress or disease conditions.
- Alert farmers via Coqui-generated voice messages.
- Recommend optimal irrigation or crop protection.



```
Microsoft Windows [version 10.0.19045.6332]
(c) Microsoft Corporation. Tous droits réservés.

C:\Users\user>ollama ps
NAME            ID              SIZE     PROCESSOR    CONTEXT    UNTIL
gpt-oss:120b    f7f8e2f8f4e0    69 GB    100% GPU     8192       Forever

C:\Users\user>ollama list
NAME                                ID              SIZE      MODIFIED
qwen3:30b-a3b-instruct-2507-fp16    c699578934a3    61 GB     4 weeks ago
gpt-oss:120b                        f7f8e2f8f4e0    65 GB     4 weeks ago
linux6200/bge-reranker-v2-m3:latest abf5c6d8bc56    1.2 GB    3 months ago
bge-m3:latest                       790764642607    1.2 GB    3 months ago
llama4:latest                       bf31604e25c2    67 GB     3 months ago

C:\Users\user>
```

Figure 6: Predictive alert loop with GPT-OSS:120B.

6 Class Diagram

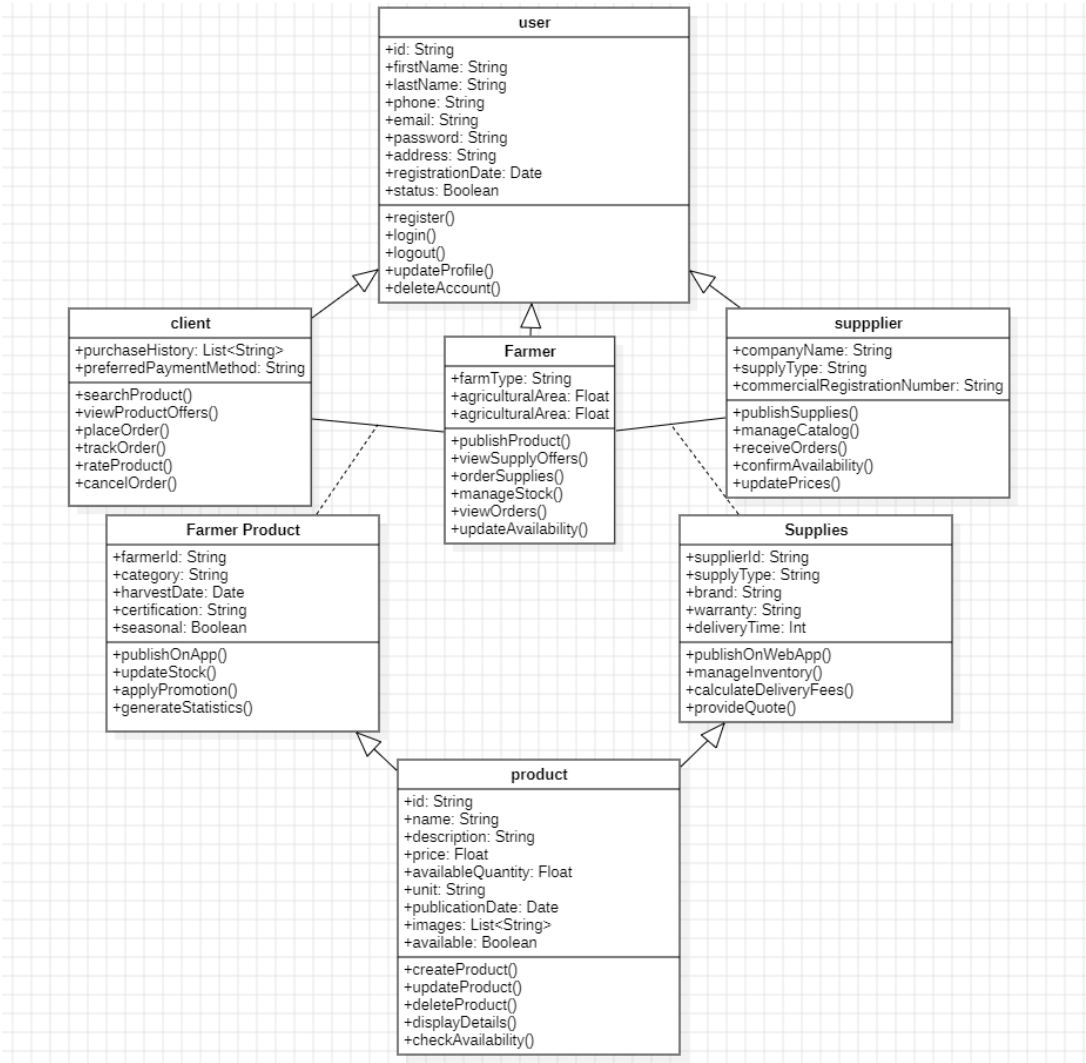


Figure 7: Mahbouba class diagram showing user, farmer, buyer, supplier, and product relationships.

7 MLOps Lifecycle with GitHub Actions

Mahbouba's machine learning lifecycle follows best DevOps practices, integrating data tracking, model deployment, and CI/CD automation.

- **DVC Pipelines:** Each dataset, feature extraction, and model training step is tracked in Git, ensuring reproducibility.
- **GitHub Actions:** Every commit triggers an automated pipeline that:
  1. Runs unit tests on AI service containers.
  2. Builds Docker images and pushes them to Azure Container Registry (ACR).
  3. Deploys updated containers to Azure Container Instances (ACI).
  4. Performs post-deployment health checks through NGINX endpoints.

- **Monitoring:** Log data collected via Azure Application Insights for latency, accuracy, and uptime analytics.

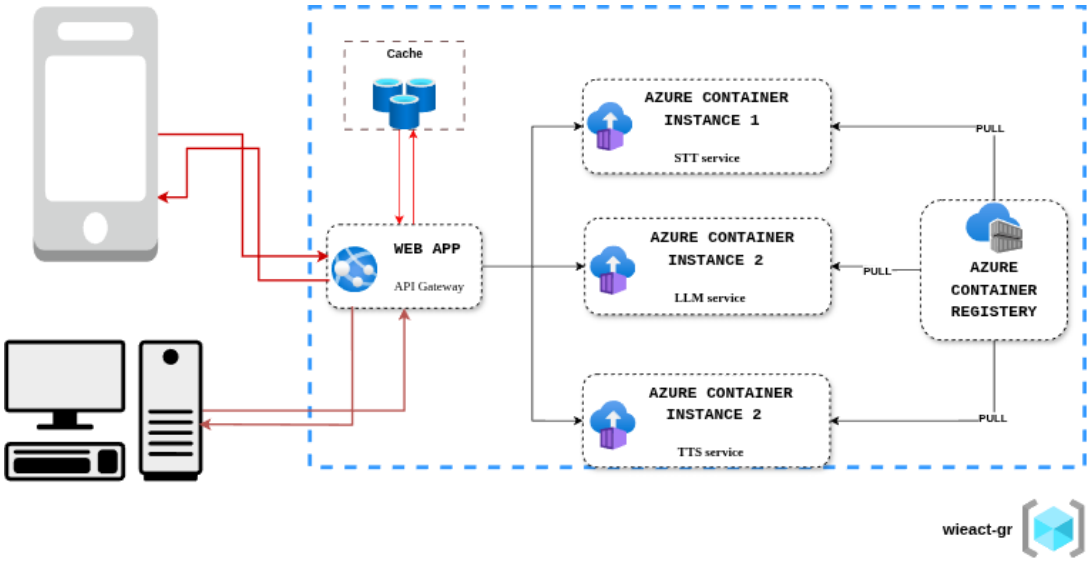


Figure 8: MLOps workflow with GitHub Actions automating CI/CD from Git to Azure.

8 Achievements

- Whisper.cpp STT, Coqui TTS, and Labess-7B fully containerized on Azure.
- NGINX Gateway and Redis caching deliver sub-second responses.
- GitHub Actions ensure continuous integration and deployment.
- DVC enables transparent versioning of models and datasets.
- IoT Meteo Station integrated with GPT-OSS:120B prediction.
- Dual database architecture: PostgreSQL for backend, Firestore for web.

9 Conclusion

Mahbouba unites AI, IoT, and automation to empower Tunisian women farmers. Through GitHub Actions, DVC, and Azure integration, the platform achieves real-time learning, prediction, and reliability — proving that socially driven MLOps solutions can drive sustainable agricultural innovation.
For full project implementation and source code, visit: `https://github.com/EmnaMnejja6/mahbouba.git`